Fiscal Year:	FY 2014	**Task Last Updated:**	FY 06/14/2014
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	jzk@fit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	336-334-5241
Organization Name:	University of North Carolina-Greensboro		
PI Address 1:	College of Arts & Sciences, 105 Foust Building		
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City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	09/11/2012	**End Date:**	12/31/2016
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 12/31/2016, per NSSC information (Ed., 6/15/15)		
Key Personnel Changes/Previous PI:	(June 2014 report): We have a new post-doctoral associate on the project--Dr. Josh Vandenbrink.		
COI Name (Institution):	Edelmann, Richard (Miami University)		
Grant/Contract No.:	NNX12AO65G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[ED. NOTE 9/28/2012: Continuation of "Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants"--grant # NNX10AM86G, per PI move from Miami-Ohio to University of Mississippi. See that project for previous reports] The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the ISS with the European Modular Cultivation System (EMCS) using the model plant Arabidopsis. In this new proposed project, we plan to confirm and extend our discovery during the recent ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Our hypothesis is that positive red-light-sensing, which was known in older plant lineages, is masked by normal 1-g conditions in more recently evolved lineages. We also propose to investigate fundamental interactions among red and blue light signaling pathways and the gravity sensing mechanisms. The ISS is the only laboratory facility where it is possible to reliably and simultaneously vary light and gravity stimuli for the proposed experiments. These experiments can be accomplished with telemetric science and therefore fit with the extremely limited down mass capabilities of current spaceflight opportunities. The EMCS is an automated facility and can be operated largely with telemetry. Experimental containers will launched via Space-X, and during an experimental run, images of seedling growth responses can be downlinked to Earth. A significant advantage of this proposed project is that we will use existing flight hardware that has been flown successfully and allowed for growth of Arabidopsis seedlings in the EMCS. This project is also relevant to the recent emphasis at NASA to study plant growth and development at fractional g-levels such as those found on the moon and Mars. Improved knowledge of the basic mechanistic processes that will be the focus of this project is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We plan to study the effects of gravity and light on basic cellular mechanisms in plants, so this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology. Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our space flight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture. In addition, improved knowledge of light and gravity systems in plants will, in the long term, allow us to develop better crop plants for use on Earth.
Task Progress:	Major milestones: 1. March 1, 2013; KSC: Launch of Space-X2 with SG-1. 2. May 2013; ISS: Completion of four experimental runs of SG-1. Images were downlinked. Seedlings were frozen in MELFI and stored on ISS until the Space X-3 mission. 3. June 2013; Madrid, Spain; Working Group for SG. 4. September 2013; Presentation of SG overview at the ELGRA meeting in Vatican City. 5. October 2013; Trondheim, Norway, Schedule Test for SG-2. 6. October 2013; Presentation of the preliminary results of SG-1 at the ASGSR meeting in Orlando. 7. December 2013; Trondheim, Norway, Operations Verification Test (OVT) for SG-2. 8. Spring 2014; PI lab; Continued image analysis of results from SG-1. 9. March-June 2014; PI lab; Propagation of plants and development of seed stocks for the SG-2 experiment. 10. April 2014; KSC; Launch of Space-X3. 11. May 2014; Return of Space-X3 with the frozen seedlings from SG-1. 12. June 2014; ARC; SG-1 samples are retrieved from hardware and are prepared for shipment to Javier Medina's lab in Spain.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Kiss JZ, Millar KD, Edelmann RE. "Operations of an experiment on plant tropisms on the International Space Station (Seedling Growth-1)." 29th Annual Meeting of the American Society for Gravitational and Space Research, Orlando, FL, November 3-8, 2013. 29th Annual Meeting of the American Society for Gravitational and Space Research, Orlando, FL, November 3-8, 2013. Program and abstracts, p. 93. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Abstracts for Journals and Proceedings	Millar KD, Sindelar T, Kiss JZ. "Red light effects on blue-light-based phototropism in roots and hypocotyls." 29th Annual Meeting of the American Society for Gravitational and Space Research, Orlando, FL, November 3-8, 2013. 29th Annual Meeting of the American Society for Gravitational and Space Research, Orlando, FL, November 3-8, 2013. Program and abstracts, p. 125. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013

Abstracts for Journals and Proceedings	Correll MJ, Pyle TP, Millar KD, Sun H, Yao J, Edelmann RE, Kiss JZ. "Transcriptome analyses of Arabidopsis thaliana seedlings grown in space: implications for gravity-responsive genes." 29th Annual Meeting of the American Society for Gravitational and Space Research, Orlando, FL, November 3-8, 2013. 29th Annual Meeting of the American Society for Gravitational and Space Research, Orlando, FL, November 3-8, 2013. Program and abstracts, p. 69. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Articles in Other Journals or Periodicals	Kiss JZ. "Book review of 'Wheels Stop: The Tragedies and Triumphs of the Space Shuttle Program, 1986-2011.' by Rick Houston." Choice. 2014 Jun;51:5573. , Jun-2014
Articles in Other Journals or Periodicals	Kiss JZ. "Book review of 'Inventing the American Astronaut.' by Matthew Hersch." Choice. 2013 Jun;50(10):5571. , Jun-2013
Articles in Peer-reviewed Journals	Kiss JZ. "Plant biology in reduced gravity on the Moon and Mars." Plant Biol (Stuttg). 2014 Jan;16 Suppl 1:12-7. Epub 2013 Jul 25. http://dx.doi.org/10.1111/plb.12031 ; PubMed PMID: 23889757 , Jan-2014
Articles in Peer-reviewed Journals	Kiss JZ, Aanes G Schiefloe M, Coelho LH, Millar KD, Edelmann RE. "Changes in operational procedures to improve spaceflight experiments in plant biology in the European Modular Cultivation System." Advances in Space Research. 2014 Mar;53(5):818–27. http://dx.doi.org/10.1016/j.asr.2013.12.024 , Mar-2014
Significant Media Coverage	Ames Research Center. "Article on experiments on ISS, including PI's experiment: 'SpaceX Dragon Spacecraft with NASA, Ames Cargo Set for Return to Earth.' " SpaceRef site, May 16, 2014. http://www.spaceref.com/news/viewpr.html?pid=43242 , May-2014